HeartSight: Project Brief

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Overview

HeartSight is a web app that predicts a person's coronary artery disease (CAD) likelihood and 10-year cardiac risk from clinical and physiological inputs, then paints those predictions onto an interactive 3D human body and heart. Tagline: **Beat the odds.**

- **The problem:** risk scores are numbers on a page. People don't act on "23%", and clinicians can't easily show patients which factors drive it.
- **The idea:** show the risk on the anatomy, explain it factor by factor, and let the user change habits in a what-if simulator and watch the heart recover.
- **The pitch in one line:** a digital twin of your heart that turns a prediction into motivation.
- **Target track:** Track A, Cardiovascular Risk Visualization & Prediction (Multimodal AI Hackathon 2026).

Challenge alignment

Every phrase in the Track A statement maps to a visible feature, so judges can tick each one off during the demo. The official one-line challenge is: *build an interactive 3D visualization system that predicts coronary artery disease and overall cardiac risk from patient physiological and clinical data, mapping predictions onto an interactive 3D human anatomical model.*

Challenge phrase	HeartSight feature
Interactive 3D visualization	Orbit, zoom and click on a torso-to-heart scene in the browser
Predicts coronary artery disease	Model 1: probability that CAD is present
Overall cardiac risk	Model 2: 10-year cardiovascular risk, with a range
Physiological data	Resting ECG result, max heart rate, exercise angina, ST depression, blood pressure

Challenge phrase	HeartSight feature
Clinical data	Age, sex, cholesterol, glucose, smoking, diabetes, family history
Mapping onto a 3D anatomical model	Risk and top drivers shown on the heart and coronary arteries inside a full torso
Multimodal (event theme)	Lab-report upload with auto-extracted values (stretch goal), plus 3D, numbers and text together

The full Track A page was not available when this brief was written. Read it for required deliverables, datasets and judging criteria, and adjust this table to match.

User experience

The app is one continuous journey in six screens. The 3D scene stays on screen the whole time, and the panels change around it.

1. **Landing.** Dark, cinematic hero with a slowly beating heart. One line: "See your heart's future. Beat the odds." One button: Start.
2. **Intake.** A stepped, conversational panel instead of one long form: basics, vitals, labs, lifestyle, exercise test results. Each field has a plain-language hint and a sensible default. A "load example patient" button fills it instantly for demos.
3. **Reveal.** The camera flies from the full torso into the heart. The coronary arteries (left main, LAD, LCX, RCA) glow from teal to amber to red. The heartbeat speeds up with risk. Two headline numbers appear: CAD likelihood and 10-year risk, each with a range.
4. **Explain.** Click any artery to see its territory, the top three factors pushing risk up or down, and a one-sentence plain-language meaning. A bar chart shows each factor's contribution.
5. **What-if simulator.** Sliders and toggles for quit smoking, lower systolic BP, lower LDL, start a statin, exercise more. Risk numbers animate down and the arteries visibly return toward teal in real time.
6. **Clinician view.** A toggle that collapses everything into a compact one-page summary with an export button.

Every screen carries a short banner: *Educational tool, not a medical diagnosis.*

Features and scope

With two days, the MVP must be small and polished. Anything below the line is written up as future work, not half-built.

Priority	Feature	Why it matters
Must	Torso-to-heart 3D scene with risk-colored coronary arteries	The core of the challenge
Must	CAD likelihood model with SHAP explanations	Prediction plus the "why"
Must	Overall 10-year risk (ASCVD equations or a second model)	Second required output
Must	What-if simulator with animated risk drop	The memorable demo moment
Must	"About the model" page: AUROC, calibration, limits	Credibility with judges
Must	Live deployed link and 2-minute demo video	Submission requirement in most hackathons
Should	Example patient personas (high, medium, low risk)	Fast, reliable demo
Should	Mobile-friendly layout and color-blind-safe palette	Polish and accessibility
Could	Lab-report PDF upload with value extraction	Multimodal angle
Could	ECG image upload	Multimodal angle
Could	Heartbeat sound and clinician PDF export	Delight
Won't (future work)	Real per-artery blockage detection from CT or angiography	Needs imaging data; say so openly

Wow features

Five tier 1 features fit in the second morning and carry the demo; tier 2 happens only if you are ahead. Each one reuses the existing model or 3D scene, so none needs new data. Hour estimates are rough guesses.

Tier 1: build these

Feature	What the judge sees	How it works	Rough cost
Heart age	"Your heart behaves like a 61-year-old's" next to the real age	Find the age at which a person with ideal risk-factor values has the same predicted risk as the patient	1 hour
Future you: split view	Two hearts side by side, "do nothing" versus "change habits", with a 10-year risk for each	Do nothing re-runs the model with age plus 10 and the same habits; change habits uses the what-if inputs. Label it a projection	2 hours
Living arteries	Blood-flow particles move through the arteries, slower and with a visibly narrower channel as risk rises	Particle speed and tube radius scale with each artery's risk score. Label it illustrative, not a measurement	2 to 3 hours
Share card	A downloadable image with heart age, risk and the disclaimer	Drawn in the browser, so nothing is uploaded	1 hour
Persona gallery	Three one-click patients: low, medium, high risk	Stored example inputs that fill the form	30 minutes

Tier 2: only if tier 1 is done and deployed

Feature	Why it helps	Watch out for
Ask HeartSight: a chat that explains the result in plain words	Makes the explanations conversational	Keep the API key on the backend; feed it only the model outputs; refuse to diagnose

Feature	Why it helps	Watch out for
Lab-report upload (photo or PDF) that fills the form	The strongest fit to the "multimodal" theme	Always show the extracted values for the user to confirm before predicting
Tamil and Hindi toggle for the summary and explanations	A real accessibility win and a memorable touch	Have a native speaker check the strings

Skip these

- Detecting blockages in a specific artery from form data. It is not possible and would hurt your credibility.
- Heart rate from a webcam. Accuracy claims are hard to defend in two days.
- Deep-learning ECG analysis. Too big for the time left.

Why these five: a judge understands each in a few seconds, each one reuses what already exists, and together they turn the demo from a prediction into a story with a before and after.

Data and models

Two models answer two different questions, and the brief keeps them separate on screen.

Model 1: CAD likelihood (is disease present?)

- **Data:** UCI Heart Disease, Cleveland subset (about 300 patients). The label comes from angiography. Published work on this dataset typically reports an AUROC around 0.9; report your own measured number, not this one.
- **Inputs:** age, sex, chest pain type, resting blood pressure, cholesterol, fasting glucose, resting ECG, max heart rate, exercise angina, ST depression, ST slope.
- **Leave out** the fluoroscopy vessel count and thalassemia test. A user cannot self-report them, and they leak information from the diagnostic work-up.
- **Algorithm:** start with logistic regression as a baseline, then gradient boosting. Keep whichever has better cross-validated AUROC and calibration. On 300 rows, simple often wins.
- **Validation:** 5-fold repeated cross-validation, never a single split. Report AUROC, calibration curve and Brier score.

Model 2: overall 10-year cardiovascular risk

- **Option A (fast and defensible):** the ASCVD Pooled Cohort Equations, implemented as published. Inputs: age, sex, total and HDL cholesterol, systolic BP, BP treatment, diabetes, smoking. Valid for ages 40 to 79.
- **Caveat to state on screen:** these equations were derived mainly in US cohorts and are reported to underestimate risk in South Asian populations. Label the output as an estimate and cite this in the model page.
- **Option B (if time allows):** a second gradient-boosting model on a larger public cohort, compared side by side with Option A.

Explainability and uncertainty

- **SHAP values** for every prediction. They drive the factor bars and the artery coloring.
- **Risk range, not a point:** use bootstrap or cross-validated spread to show, for example, "18% (range 12 to 26%)".
- **What-if simulator:** re-run the model with the changed inputs. Do not hard-code the effect of a statin or quitting smoking; if you need a fixed effect size, cite its source on the model page.

Honest limits (put these on the About page)

- Tabular data cannot locate disease in a specific artery. The per-artery colors are an illustrative mapping, not a detection.
- Small training set; results will not generalize to every population.
- Not a medical device and not for diagnosis.

3D design

The scene is built with React Three Fiber and has two levels: the whole body for context, and the heart for detail.

- **Models:** a free torso or body model with the heart visible, or a stylized heart plus tube-shaped coronary arteries you build yourself. Check each model's license (Creative Commons attribution is common) and credit it in the footer.
- **Size budget:** keep all models under about 5 MB total, compressed (glTF with Draco). Test on a mid-range phone early.
- **Camera:** an animated fly-in from torso to heart after the prediction runs, with orbit and zoom afterwards.

- **Risk coloring:** a custom shader or material color blends teal, amber and red from a 0 to 1 risk value per artery. Use a color-blind-safe palette and always show the number too.
- **Heartbeat:** animation speed scales with the overall risk, from a calm rate to a visibly faster one.
- **Picking:** clicking an artery opens its detail panel; hovering shows its name.

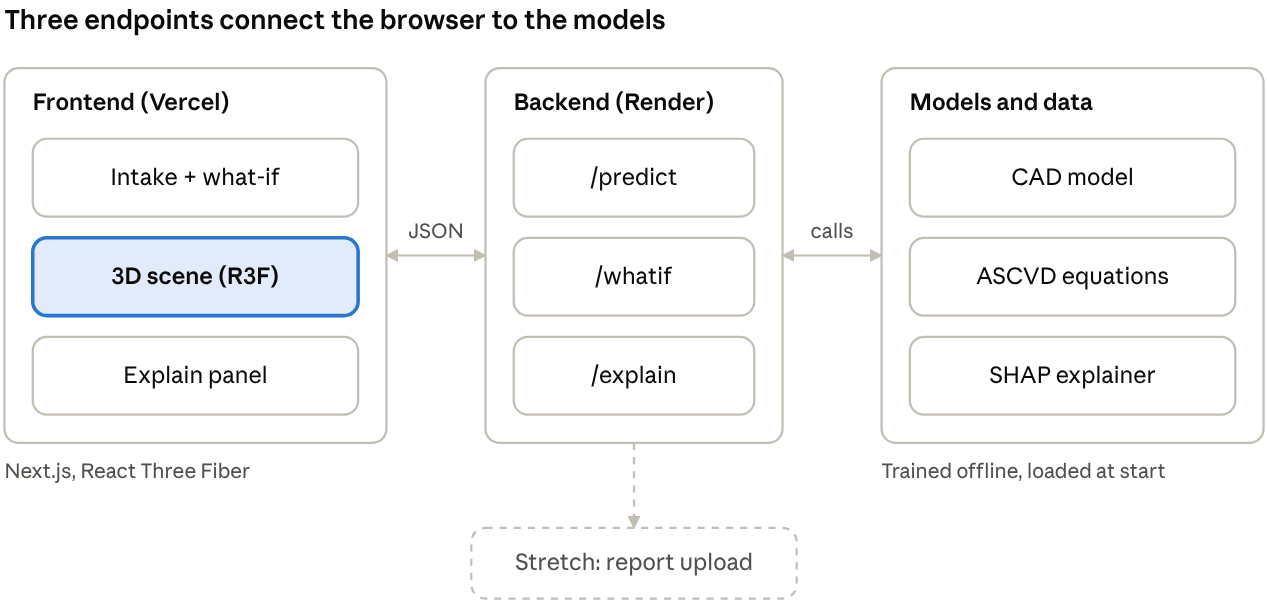
Artery mapping (illustrative, documented on the About page)

Each artery gets a score built from the SHAP contributions of the factors clinically linked to it. The weights are heuristics you write down openly, not learned from artery-level data.

Artery	Territory	Factors weighted most (heuristic)
Left main	Feeds both the LAD and LCX	Overall model risk (highest of the others)
LAD	Front wall of the left ventricle	Blood pressure, age, diabetes
LCX	Side and back of the left ventricle	Cholesterol, age
RCA	Right side and bottom of the heart	Smoking, glucose, ECG findings

Check these weights with a clinician or a cardiology source before you present them. If you can't, keep the mapping simpler: color all arteries by overall risk and use the factors only in the side panel.

Architecture and tech stack



architecture · 3 layers, 3 endpoints

The browser sends inputs to three endpoints; the backend runs the models and returns risks and SHAP values, and the 3D scene only renders the result. The dashed box is the stretch goal.

Layer	Choice	Why
Frontend	Next.js, React, Tailwind, Framer Motion	Fast to build, one-click deploy
3D	React Three Fiber, drei	Three.js with React state
Charts	Recharts	SHAP bars, calibration plot
Backend	FastAPI	Python ML libraries, automatic API docs
Machine learning	scikit-learn, XGBoost or LightGBM, SHAP	Standard and well documented
Report extraction (stretch)	An LLM API or OCR library	Multimodal input
Hosting	Vercel (frontend), Render or Railway (backend)	Public link judges can open

Free backend tiers can sleep when idle. Open the app a few minutes before the demo, or ping the backend on page load.

Two-day build plan

The rule for the two days: a working end-to-end version with a public link by the end of day 1, then wow features and polish on day 2, with a feature freeze at midday. Dates assume submission closes on Thursday 15 October; confirm the exact deadline and time zone first.

When	Goal	Done when
Tonight (Tue 13 Oct)	Register, read the full Track A page and rules, pick the 3D model, scaffold both projects, train the baseline CAD model	Registered; both projects run; baseline cross-validated AUROC saved
Day 1 (Wed 14 Oct)	Finish both models and the API, 3D scene with colored arteries, what-if simulator, deploy a thin version	A persona goes in, arteries change, and a public link exists, even if ugly

When	Goal	Done when
Day 2 (Thu 15 Oct)	Wow pack, About page, polish, phone test, README, demo video, submit	Live link works on a phone; video uploaded; submission sent hours before the deadline

Day 1 in more detail (Wed 14 Oct)

1. Morning: finish training (logistic regression and gradient boosting), save the best model, compute SHAP.
2. Midday: write `/predict`, `/whatif` and `/explain` in FastAPI and implement the ASCVD equations.
3. Afternoon: build the R3F scene with four colored arteries, wire the form to the API, add basic what-if sliders.
4. Evening: run three example patients end to end and deploy a thin version to a public link.

Day 2 in more detail (Thu 15 Oct, submission day)

1. Morning: the tier 1 wow features (heart age, future-you split view, blood flow and narrowing, share card). Freeze features at midday.
2. Midday: About the model page, landing page, fly-in camera, mobile layout, color-blind-safe palette.
3. Afternoon: final deploy, phone and second-browser test, README, record the 2-minute demo video.
4. Submit with at least three hours to spare. Tier 2 wow features only if everything above is done.

Demo script and judging pitch

The demo is a story about one person, not a tour of features. Use a persona such as Ravi, 52, a smoker with high blood pressure and high LDL. Rehearse it until it takes under two minutes.

Time	What happens on screen	What you say
0:00 to 0:15	Landing page, heart beating	"Risk scores are numbers. HeartSight makes them visible."

Time	What happens on screen	What you say
0:15 to 0:35	Load Ravi, press Predict, camera flies into the heart, heart age appears	"Ravi is 52, but his heart behaves much older. Two predictions: is disease likely present, and his 10-year risk."
0:35 to 1:00	Click the red artery; blood flow slows, narrowing shows, SHAP bars appear	"Smoking and LDL drive most of this. Every number is explained."
1:00 to 1:35	Toggle quit smoking, lower BP, start statin; split view shows do nothing versus change habits; arteries turn teal and heart age drops	"Same patient, new choices. This is the part people remember."
1:35 to 1:50	About page: AUROC, calibration, limits	"We show our accuracy and what the tool cannot do."
1:50 to 2:00	Share card and closing slide with the link	"HeartSight. Beat the odds."

How this should score

- **Technical depth:** two models, SHAP, calibration, risk ranges, a benchmark against a clinical standard.
- **Fit to the challenge:** each phrase of the statement is visible on screen (see the alignment table).
- **Design and polish:** cinematic 3D, smooth animation, works on a phone.
- **Responsibility:** the limits are stated on every screen and on the About page.

If judges ask what the weakest part is, answer first: the artery coloring is illustrative, because tabular data cannot localize disease, and real localization needs imaging.

Risks, ethics and mitigations

Risk	Likelihood	Mitigation
No good, light 3D heart model found	Medium	Build a stylized heart with tube arteries; decide by the end of tonight
3D scene is slow on phones	Medium	Compress models, cap pixel ratio, test on a phone on day 1

Risk	Likelihood	Mitigation
Overclaiming per-artery detection	High if careless	Label the coloring illustrative everywhere; explain on the About page
Small dataset gives unstable results	High	Cross-validation, risk ranges, simple models, honest reporting
Risk equations are less accurate for some groups	Known	State the South Asian under-estimation caveat; show it as an estimate
Backend sleeps or deploy fails on day 3	Medium	Deploy a thin version on day 1; keep a recorded demo as a fallback
Running out of time	High	Hold the cut list; the must-have table is the whole scope
Wow features eat the schedule	High	Core first; start tier 1 only after the end-to-end demo works on day 1 evening; freeze at midday on day 2

Ethics and privacy commitments

- The app says it is educational and not a diagnosis, on every screen.
- No patient data is stored by default. Inputs live in the browser session and are sent only to compute the prediction.
- Datasets and 3D models are credited with their licenses.
- Performance is checked across sex and age groups, and any gap is reported on the About page.

Submission checklist

The hackathon PDF lists submission guidelines and the prize pool only as headings, so check the official page for the exact required items and add them here.

- [ ] Registered for the hackathon and Track A
- [ ] Read the full Track A problem statement and submission guidelines
- [ ] Confirmed whether teams are allowed and the exact deadline and time zone
- [ ] Dataset and 3D model licenses checked and credited

- [ ] CAD model trained, cross-validated, and metrics saved
- [ ] 10-year risk implemented and benchmarked
- [ ] 3D scene with four colored arteries and click-to-explain
- [ ] What-if simulator working
- [ ] About the model page with AUROC, calibration and limits
- [ ] Disclaimer on every screen
- [ ] Deployed, with a public link tested on a phone
- [ ] README with setup steps, screenshots and architecture diagram
- [ ] 2-minute demo video recorded and uploaded
- [ ] Submission sent with hours to spare